

FEDERAL UNIVERSITY LOKOJA
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION 2021/22 SESSION
PHY 803: QUANTUM MECHANICS
Time: 3 Hrs 18th October, 2023 3 CREDITS.

Instruction: Answer any FOUR questions.

Planck constant $h = 6.626 \times 10^{-34}$ Js, $1\text{eV} = 1.6 \times 10^{-19}$ J,
Electronic mass $m_e = 9.11 \times 10^{-31}$ Kg, Mass of proton $m_p = 1.672 \times 10^{-27}$ Kg
Mass of Neutron $m_n = 1.678 \times 10^{-27}$ Kg

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}, \quad \int_{-\infty}^{\infty} x^{(2n)} e^{-\alpha x^2} dx = \frac{1 \times 3 \times 5 \times 7 \times \cdots \times (2n-1)}{2^n \alpha^n} \sqrt{\frac{\pi}{\alpha}}$$
$$\int_0^{\infty} x^{(2n+1)} e^{-\alpha x^2} dx = \frac{n!}{2\alpha^{n+1}}, \quad \int_0^{\infty} x^n e^{-x} dx = n!$$

1. (a) State Heisenberg's uncertainty relation.
(b) Using an arbitrary wavefunction or otherwise derive the commutation relations between the following quantities:
(i) the position and momentum components $\hat{X}$ and $\hat{P}_x$
(ii) the angular momentum components $\hat{L}_x$ and $\hat{L}_y$
(c) Explain briefly the physical implication of the relations in question (1b).

2. (a) What does the term *degenerate state* mean?
(b) Determine the eigenvalues and the normalized eigenvectors of an operator $\hat{O}$ given by

$$\hat{O} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

- (c) Verify whether the eigenvectors form an orthonormal and complete basis.
3. (a) Distinguish between Unitary and Hermitian operators.
(b) A free particle in one dimension ($H = p^2/2m$) is in the state $|1\rangle$ whose normalized wave function is $\psi_1(x) = (\frac{\alpha}{\pi})^{1/4} \exp(-\alpha x^2/2)$. Calculate the expectation values of p^2 .
(c) Consider the matrices

$$\hat{A} = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 1 & -i \\ 0 & i & -1 \end{bmatrix} \quad \hat{B} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 2i & 0 \\ i & 0 & -5i \end{bmatrix}$$

- (i) Proof the Hermiticity of operators $\hat{A}$ and $\hat{B}$, ii) Show that $\hat{A}$ and $\hat{B}$ do not commute.
4. (a) What is a stationary state of quantum system?
(b) The Hamiltonian $\hat{H}$ and the initial state $|\psi_i\rangle$ of a physical system are respectively given by

$$\hat{H} = \epsilon \begin{bmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad |\psi_i\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} 1-i \\ 1-i \\ 1 \end{bmatrix},$$

where ε has the dimension of energy.

(i) Evaluate the energy eigenvalues. (ii) Calculate the respective probabilities of the eigenvalues and hence the expectation value $\langle \hat{H} \rangle$. (iii) Determine the expectation value of the Hamiltonian using the matrix mechanics.

5. (a) Given a quantum state $\Psi(\vec{r}) = \sum_i c_i \phi_i(\vec{r})$ expressed as a superposition over an orthonormal basis set $\{\phi_i(\vec{r})\}$. Deduce the expression for each of the following:
- (i) The expansion coefficients. (ii) Completeness of the expansion coefficients.
 - (iii) The closure relation for the set $\{\phi_i(\vec{r})\}$.
- (b) A 3-dimensional subspace of the Hilbert space contains an orthonormal basis $\{|\varphi_1\rangle, |\varphi_2\rangle, |\varphi_3\rangle\}$. If the kets $|\alpha\rangle = i|\varphi_1\rangle - 2|\varphi_2\rangle - i|\varphi_3\rangle$ and $|\beta\rangle = i|\varphi_1\rangle + 2|\varphi_3\rangle$, are members of the space. Evaluate the following quantities:
- (i) The scalar product $\langle \alpha | \beta \rangle$, (ii) The operator $|\alpha\rangle\langle \beta|$, (iii) The projection operator onto $|\alpha\rangle$.
6. (a) (i) Derive the generalized uncertainty principle

$$\sigma_A \sigma_B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

between non commuting linear operators $\hat{A}$ and $\hat{B}$.

(ii) Use the result obtained in a(i) to verify the uncertainty relations between $\hat{X}$ and $\hat{P}_x$, and $\hat{L}_x$ and $\hat{L}_y$.

- (b) Show that the eigenvalues of an Hermitian operators are real and the eigenvectors corresponding to different eigenvalues are orthogonal.